

EdgeMonitor Gateway Requirements

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Project: EdgeMonitor Gateway Pilot

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1. Product Context

EdgeMonitor Gateway is a pilot industrial IoT gateway used to collect telemetry from factory machines, buffer data locally during cloud outages, and forward validated records to a cloud analytics platform.

The gateway will be demonstrated on a simulated production line with three device types:

- temperature controller using Modbus TCP;
- vibration sensor using MQTT;
- energy meter using HTTP JSON API.

The project team needs a test plan that covers functional behavior, reliability during outages, security access control, and traceability from requirements to tests.

2. User Roles

Plant Operator

The plant operator monitors machine status and acknowledges alerts. The operator shall not modify device configuration.

Maintenance Engineer

The maintenance engineer registers devices, edits signal mappings, and validates telemetry collection.

Site Administrator

The site administrator manages users, certificates, configuration rollback, and diagnostic exports.

3. Functional Requirements

REQ-EM-001 Device Registration

The gateway shall allow a maintenance engineer to register a new device with a unique device ID, protocol type, host, port, polling interval, and signal map.

Acceptance notes:

- duplicate device IDs shall be rejected;
- unsupported protocol types shall be rejected;
- successful registration shall create an audit event.

REQ-EM-002 Telemetry Normalization

The gateway shall normalize all telemetry records into a common schema containing:

- event_id;
- device_id;
- signal_id;
- original_timestamp;
- ingestion_timestamp;
- value;
- unit;

- quality;
- source_protocol.

REQ-EM-003 Polling Interval Boundary

The gateway shall reject polling intervals below 1 second.

The default polling interval shall be 10 seconds when no polling interval is supplied.

REQ-EM-004 Local Buffering

If the cloud ingestion endpoint is unavailable, the gateway shall buffer telemetry locally for at least 6 hours at a rate of 500 measurements per minute.

REQ-EM-005 Chronological Replay

When cloud connectivity is restored, the gateway shall forward buffered telemetry in chronological order.

REQ-EM-006 Duplicate Prevention

The gateway shall not forward duplicate telemetry records after a transient cloud timeout.

Each telemetry record shall use a stable event_id for idempotency.

REQ-EM-007 Device Health State

The gateway shall update a device health state to disconnected within 30 seconds after communication failure is detected.

Allowed health states are:

- healthy;
- degraded;
- disconnected;
- configuration_error.

REQ-EM-008 Buffer Capacity Alert

The gateway shall generate an alert when local buffer usage exceeds 80 percent for more than 60 seconds.

REQ-EM-009 Configuration Versioning

Every device configuration update shall create an immutable configuration version.

REQ-EM-010 Rollback

The site administrator shall be able to roll back to the previous valid configuration version.

Rollback shall not delete the failed configuration version from the audit history.

4. Security Requirements

REQ-EM-011 Authenticated API Access

All API endpoints except /health/live shall require authentication.

Unauthenticated requests shall return HTTP 401.

REQ-EM-012 Role-Based Authorization

The system shall enforce role-based authorization:

- plant_operator may read device status and acknowledge alerts;
- maintenance_engineer may register devices and edit signal maps;
- site_admin may manage users, rollback configuration, and export diagnostics.

REQ-EM-013 Diagnostic Redaction

Diagnostic exports shall redact passwords, API tokens, private keys, bearer tokens, and cloud credentials.

REQ-EM-014 TLS Validation

The cloud forwarder shall validate the cloud endpoint certificate chain before sending telemetry.

If TLS validation fails, telemetry shall remain buffered locally.

5. Non-Functional Requirements

REQ-EM-015 Startup Readiness

After power-on, the gateway shall be ready to collect telemetry within 90 seconds.

REQ-EM-016 Nominal Load

The gateway shall process at least 500 measurements per minute with CPU usage below 70 percent on the reference hardware.

REQ-EM-017 Audit Retention

The system shall retain audit events for at least 180 days.

6. Out of Scope

The following items are out of scope for the pilot:

- machine learning inference at the edge;
- mobile application;
- multi-site orchestration;
- high-frequency vibration streaming above 1 kHz.